

SSH Secure Channel



VMware

```
(kali@kali)-[~]
└─$ ssh -oHostKeyAlgorithms=ssh-rsa iotgoatuser@192.168.217.153
The authenticity of host '192.168.217.153 (192.168.217.153)' can't be established.
RSA key fingerprint is SHA256:ZNiW8hNNt0FwhkmR5nWeUrQXmYTNqn2jPAXNSVzZ1o0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.217.153' (RSA) to the list of known hosts.
iotgoatuser@192.168.217.153's password:

BusyBox v1.28.4 () built-in shell (ash)

OWASP
IoTGOAT

GitHub: https://github.com/OWASP/IoTGoat

iotgoatuser@IoTGoat:~$ ^C
iotgoatuser@IoTGoat:~$
```

Step1: 3 Way Handshake

No.	Time	Source	Destination	Protocol	Length	Info
58	43.61348...	192.168.217.154	192.168.217.153	TCP	74	45772 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=4293026543 TSecr=
59	43.61376...	192.168.217.153	192.168.217.154	TCP	74	22 → 45772 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSval=25152
60	43.61379...	192.168.217.154	192.168.217.153	TCP	66	45772 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=4293026544 TSecr=2515268612

Step2: Identification exchange

No.	Time	Source	Destination	Protocol	Length	Info
61	43.61404...	192.168.217.154	192.168.217.153	SSHv2	98	Client: Protocol (SSH-2.0-OpenSSH_9.9p1 Debian-3)
63	43.61453...	192.168.217.153	192.168.217.154	SSHv2	372	Server: Protocol (SSH-2.0-dropbear), Key Exchange Init
65	43.61466...	192.168.217.154	192.168.217.153	SSHv2	1178	Client: Key Exchange Init
67	43.66013...	192.168.217.154	192.168.217.153	SSHv2	114	Client: Elliptic Curve Diffie-Hellman Key Exchange Init
69	43.69282...	192.168.217.153	192.168.217.154	SSHv2	690	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys
70	43.70625...	192.168.217.154	192.168.217.153	SSHv2	82	Client: New Keys
72	43.75036...	192.168.217.154	192.168.217.153	SSHv2	130	Client: Encrypted packet (len=64)

Step3,4: Algorithm negotiation

No.	Time	Source	Destination	Protocol	Length	Info
61	43.61404...	192.168.217.154	192.168.217.153	SSHv2	98	Client: Protocol (SSH-2.0-OpenSSH_9.9p1 Debian-3)
63	43.61453...	192.168.217.153	192.168.217.154	SSHv2	372	Server: Protocol (SSH-2.0-dropbear), Key Exchange Init
65	43.61466...	192.168.217.154	192.168.217.153	SSHv2	1178	Client: Key Exchange Init
67	43.66013...	192.168.217.154	192.168.217.153	SSHv2	114	Client: Elliptic Curve Diffie-Hellman Key Exchange Init
69	43.69282...	192.168.217.153	192.168.217.154	SSHv2	690	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys
70	43.70625...	192.168.217.154	192.168.217.153	SSHv2	82	Client: New Keys
72	43.75036...	192.168.217.154	192.168.217.153	SSHv2	130	Client: Encrypted packet (len=64)

```
SSH Protocol
  Protocol: SSH-2.0-dropbear
  SSH Version 2 (encryption:aes128-ctr mac:hmac-sha2-256 compression:none)
    Packet Length: 284
    Padding Length: 9
    Key Exchange (method:curve25519-sha256@libssh.org)
      Message Code: Key Exchange Init (20)
      Algorithms
        Cookie: e9 78 fa 5c eb c2 72 f5 18 93 a1 06 68 6e 22 11
        kex_algorithms length: 109
        kex_algorithms string: curve25519-sha256@libssh.org,diffie-hellman-group14-sha1,
        server_host_key_algorithms length: 7
        server_host_key_algorithms string: ssh-rsa
        encryption_algorithms_client_to_server length: 21
        encryption_algorithms_client_to_server string: aes128-ctr,aes256-ctr
        encryption_algorithms_server_to_client length: 21
        encryption_algorithms_server_to_client string: aes128-ctr,aes256-ctr
        mac_algorithms_client_to_server length: 23
        mac_algorithms_client_to_server string: hmac-sha1,hmac-sha2-256
        mac_algorithms_server_to_client length: 23
        mac_algorithms_server_to_client string: hmac-sha1,hmac-sha2-256
        compression_algorithms_client_to_server length: 4
        compression_algorithms_client_to_server string: none
```

Step3,4: Key exchange from client

```
SSH Protocol
  SSH Version 2 (encryption:aes128-ctr mac:hmac-sha2-256 compression:none)
    Packet Length: 1108
    Padding Length: 4
    Key Exchange (method:curve25519-sha256@libssh.org)
      Message Code: Key Exchange Init (20)
      Algorithms
        Cookie: 89 4f 8b 4e 2d b2 ac f1 be f8 ec e2 e4 ff 12 b9
        kex_algorithms length: 350
        kex_algorithms string [...]: sntrup761x25519-sha512,sntrup761x25519-sha512@openssh.com
        server_host_key_algorithms length: 7
        server_host_key_algorithms string: ssh-rsa
        encryption_algorithms_client_to_server length: 108
        encryption_algorithms_client_to_server string: chacha20-poly1305@openssh.com,aes
        encryption_algorithms_server_to_client length: 108
        encryption_algorithms_server_to_client string: chacha20-poly1305@openssh.com,aes
        mac_algorithms_client_to_server length: 213
        mac_algorithms_client_to_server string [...]: umac-64-etm@openssh.com,umac-128-etm
        mac_algorithms_server_to_client length: 213
        mac_algorithms_server_to_client string [...]: umac-64-etm@openssh.com,umac-128-etm
        compression_algorithms_client_to_server length: 21
        compression_algorithms_client_to_server string: none,zlib@openssh.com
        compression_algorithms_server_to_client length: 21
```

Step5,6,7: Server Generate the Hash

No.	Time	Source	Destination	Protocol	Length	Info
61	43.61404...	192.168.217.154	192.168.217.153	SSHv2	98	Client: Protocol (SSH-2.0-OpenSSH_9.9p1 Debian-3)
63	43.61453...	192.168.217.153	192.168.217.154	SSHv2	372	Server: Protocol (SSH-2.0-dropbear), Key Exchange Init
65	43.61466...	192.168.217.154	192.168.217.153	SSHv2	1178	Client: Key Exchange Init
67	43.66013...	192.168.217.154	192.168.217.153	SSHv2	114	Client: Elliptic Curve Diffie-Hellman Key Exchange Init
69	43.69282...	192.168.217.153	192.168.217.154	SSHv2	690	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply
70	43.70625...	192.168.217.154	192.168.217.153	SSHv2	82	Client: New Keys
72	43.75036...	192.168.217.154	192.168.217.153	SSHv2	130	Client: Encrypted packet (len=64)

```
SSH Protocol
  SSH Version 2 (encryption:aes128-ctr mac:hmac-sha2-256 compression:none)
    Packet Length: 44
    Padding Length: 6
    Key Exchange (method:curve25519-sha256@libssh.org)
      Message Code: Elliptic Curve Diffie-Hellman Key Exchange Init (30)
      ECDH client's ephemeral public key length: 32
      ECDH client's ephemeral public key (Q_C): eb 24 e8 3d 47 64 bf 0a b8 a4 8c fa d4 b8 c1 82 69
      Padding String: 00 00 00 00 00 00
      [Sequence number: 1]
      [Direction: client-to-server]
```

Step 5, 6, 7: Hash is Generated & Signed by its Private Key

```
SSH Protocol
  SSH Version 2 (encryption:aes128-ctr mac:hmac-sha2-256 compression:none)
    Packet Length: 604
    Padding Length: 8
    Key Exchange (method:curve25519-sha256@libssh.org)
      Message Code: Elliptic Curve Diffie-Hellman Key Exchange Reply (31)
      KEX host key (type: ssh-rsa)
        Host key length: 279
        Host key type length: 7
        Host key type: ssh-rsa
        Multi Precision Integer Length: 3
        RSA public exponent (e): 01 00 01
        Multi Precision Integer Length: 257
        RSA modulus (N) [...]: 00 c4 e1 f3 35 ad f0 a7 8a 31 98 d1 1a 74 53 f4 b6 59 ff a2 32 3d 91
        ECDH server's ephemeral public key length: 32
        ECDH server's ephemeral public key (Q_S): c8 3e f9 a1 8f b8 64 42 a6 4c 03 5a 6e 74 ef ec 5a
      KEX host signature (type: ssh-rsa)
        Host signature length: 271
        Host signature type length: 7
        Host signature type: ssh-rsa
        Multi Precision Integer Length: 256
        RSA signature [...]: 2e 66 aa f1 82 f3 1c 85 da bf 24 a4 92 aa 63 b6 ae be 9a 42 af 41 d1 d7
    Padding String: 2a e4 af d5 b2 be d9 5f
    [Sequence number: 1]
```


Step8: Generate Secret Key

No.	Time	Source	Destination	Protocol	Length	Info
61	43.61404...	192.168.217.154	192.168.217.153	SSHv2	98	Client: Protocol (SSH-2.0-OpenSSH_9.9p1 Debian-3)
63	43.61453...	192.168.217.153	192.168.217.154	SSHv2	372	Server: Protocol (SSH-2.0-dropbear), Key Exchange Init
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70	43.70625...	192.168.217.154	192.168.217.153	SSHv2	82	Client: New Keys
72	43.75036...	192.168.217.154	192.168.217.153	SSHv2	130	Client: Encrypted packet (len=64)

SSH Protocol

> SSH Version 2 (encryption:aes128-ctr mac:hmac-sha2-256 compression:none)

~ SSH Version 2 (encryption:aes128-ctr mac:hmac-sha2-256 compression:none)

Packet Length: 12

Padding Length: 10

~ Key Exchange (method:curve25519-sha256@libssh.org)

Message Code: New Keys (21)

Padding String: e1 ed 91 03 2b 2a a0 90 3e 69

[Sequence number: 2]

[Direction: **server-to-client**]

SSH Protocol

~ SSH Version 2 (encryption:aes128-ctr mac:hmac-sha2-256 compression:none)

Packet Length: 12

Padding Length: 10

~ Key Exchange (method:curve25519-sha256@libssh.org)

Message Code: New Keys (21)

Padding String: 00 00 00 00 00 00 00 00 00 00

[Sequence number: 2]

[Direction: **client-to-server**]

Time Line

